

Grade 7
Unit 12
Lesson 5 Practice

Name Answer Key
Date _____
Period _____

The dimensions of four cylinders are given in the column on the left. Match each set of dimensions with the corresponding volume of the cylinder.

1. D $r = 7$ units; $h = 4$ units

$$V = \pi \cdot 7^2 \cdot 4 = 196\pi$$

A. 12π units³

2. B $r = 3$ units; $h = 7$ units

$$V = \pi \cdot 3^2 \cdot 7 = 63\pi$$

B. 63π units³

3. A $d = 4$ units; $h = 3$ units

$$V = \pi \cdot 2^2 \cdot 3 = 12\pi$$

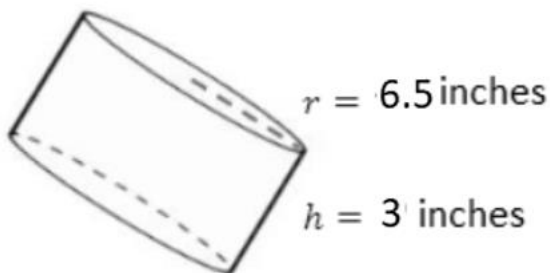
C. 144π units³

4. C $d = 12$ units; $h = 4$ units

$$V = \pi \cdot 6^2 \cdot 4 = 144\pi$$

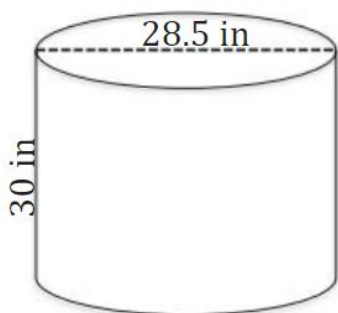
D. 196π units³

5. Calculate the volume of the cylinder. Use 3.14 for π and round to the nearest thousandths place.



$$\begin{aligned} V &= \pi r^2 h \\ V &= 3.14 \cdot 6.5^2 \cdot 3 \\ V &= 397.995 \text{ in}^3 \end{aligned}$$

6. Calculate the volume of the cylinder. Leave the volume in terms of π .

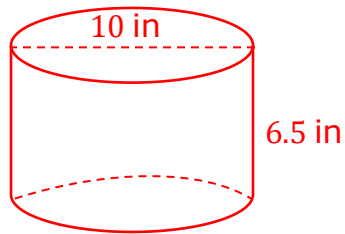


$$r = 14.25$$

$$\begin{aligned} V &= \pi r^2 h \\ V &= \pi \cdot 14.25^2 \cdot 30 \\ V &= 6,091.875\pi \text{ in}^3 \end{aligned}$$

Gunnar has a watering bowl that is shaped like a right circular cylinder. The bowl is 10 inches (in.) in diameter and has a height of 6.5 in.

7. Draw a picture of the bowl and label the dimensions.



8. How many cubic inches of water can the bowl hold? Use $\frac{22}{7}$ for π and round your answer to the nearest tenths place.

$$\begin{aligned}V &= \pi r^2 h \\V &= \frac{22}{7} \cdot 5^2 \cdot 6.5 \\V &= 510.7 \text{ in}^3\end{aligned}$$

Sammy calculated the volume of a cylinder with a diameter of 9 feet (ft.) and height of 20 ft. His calculations are shown.

$$\begin{aligned}(\pi \cdot 9^2)(20) \\(81\pi)(20) \\1620\pi\end{aligned}$$

9. Did Sammy calculate the volume correctly? If not, describe the mistake he made.

No, Sammy used the diameter in his formula instead of the radius.

10. Find the volume and leave your answer in terms of π .

$$\begin{aligned}V &= \pi r^2 h \\V &= \pi \cdot 4.5^2 \cdot 20 \\V &= 405\pi \text{ ft}^3\end{aligned}$$